

Bootstrap Interview Questions for Freshers

1) What is Bootstrap?

It is a computer programming tool with pre-written strings of code or snippets, which can be used for web development to add interactive interfaces and design the website. Its code consists of **HTML**(Hypertext Markup Language) and **CSS**(Cascading Style Sheets). We use it for designing tables, forms, typography, etc. It also has an optional javascript plugin. For mobile web applications, Bootstrap is mainly recommended.

2) What is the basic use of Bootstrap?

Bootstrap is mainly used to quickly build responsive and visually attractive websites without creating them from scratch. It delivers pre-designed elements and styles, making web development more effortless.

3) What are Bootstrap types?

Bootstrap offers two primary layout types: Fluid Layout and Fixed Layout. Fluid Layout adjusts content to screen size, while Fixed Layout maintains a consistent width.

4) What are the advantages of Bootstrap?

The following given are the advantages of Bootstrap-

- We can quickly use Bootstrap with a basic understanding of HTML and CSS.
- We have responsive CSS in Bootstrap, which can be adapted to tablets, phones, and desktops.
- Modern browsers are compatible, including Chrome, Internet Explorer, Firefox, Brave, Edge, Safari, and Opera with Bootstrap 4.

5) What is a Bootstrap Container, and how does it work?

It is a handy class that generates a central region on the page where your site content can be put. Being responsive and containing all HTML code are the advantages of Bootstrap '.container'. We use a container to pad the content inside the bootstrap container.

There are two types of containers-

- The **.container** class can create a responsive and fixed-width container.
- The **.container-fluid** class can create a full-width container that spans the entire viewport width.
- The **.container-{breakpoint}**, with width of 100% until a specified breakpoint.

6) What do the default Bootstrap text settings consist of?

In Bootstrap 4, the text is set at a default font size of 16 pixels, with a line spacing 1.5. The standard font family is "Helvetica Neue," followed by "Helvetica," "Arial," and a generic sans-serif. Moreover, <p> elements have a margin-top of 0 and a margin-bottom of 1 rem (16 pixels by default).

7) What is a lead?

In Bootstrap, developers use the "lead" class to emphasize a paragraph of text. This class improves the font size and adds spacing to the paragraph, making it stand out and capturing the reader's engagement. People often use it for training or emphasizing important text content on web pages.

8) What does the term 'flexbox' refer to in Bootstrap 4?

In Bootstrap 4, 'flexbox' refers to a layout system that helps arrange elements within a webpage flexibly and efficiently, enabling easier alignment, distribution, and content responsiveness.

9) Explain the difference between Bootstrap 4 and Bootstrap 5?

In Bootstrap 4, form elements appeared as the default style provided by web browsers. But with Bootstrap 5, these form elements come with a custom design, ensuring a consistent and unified appearance across all web browsers.

10) Explain the difference between Bootstrap 3 and Bootstrap 4?

Bootstrap 3 and Bootstrap 4 are website design tools, but they differ. Bootstrap 4 offers additional advanced features, more useful grid systems, and enhanced typography, making it better relaxed for modern web design than the older Bootstrap 3.

11) Why do we use Bootstrap for designing websites?

We use Bootstrap since it has many attractive qualities like -

- Easy to learn.
- Bootstrap supports all common browsers.
- Its code is responsive to every device.
- It contains files for coding websites for mobile devices.

12) What are the main components of Bootstrap?

The Bootstrap key components are-

- **Scaffolding**
This component provides a basic structure with link style, Grid system, and background s provided by this component.
- **CSS**
Various CSS files are contained here.
- **Layout component**
This component gives the list of layout components.
- **Customize**
You can customize your components to get get your version of the framework.
- **JavaScript plugins**
Many jQuery and JavaScript plugins are contained here.

13) What is input groups in Bootstrap?

The input-group class acts like a box to make inputs better. It adds things like symbols or buttons before or after the input field. It works well with Bootstrap 5 for form text. To put help text before the input, use ".input-group" and add the form.

14) What do you mean by Bootstrap ClassLoader?

It is a part of Java and a main parental class of ClassLoader. Named classes can be changed to binary code by class loaders.

15) Which two codes do we use for code display in Bootstrap?

There are two direct ways to display code in Bootstrap-

- We use the `<code>` element to showcase a piece of inline code.
- For displaying a code with multiple lines or a block element, we use the `<pre>` tag.

For example-

You can use the `<pre>` tag like this-

- HTML

```
<pre>
We display text in pre-tag in a fixed-width
font, and it preserves
Both spaces and
line breaks.
</pre>
```

And `<code>` tag can be used as -

- HTML

```
<code>int var = 10;</code> will be wrapped as inline.
```

16) What is a progress bar in Bootstrap?

We use the progress bar in Bootstrap to show a process's progress. In HTML element that are using `<progress>` keyword with HTML tag style we use progress bar. `<progress>` is used with CSS classes only made for the progress bar.

17) What do you mean by Bootstrap media query?

Based on viewport size or user's device size, we can change the design, positioning, moving, hiding elements/content, etc., done with media queries in Bootstrap.

- HTML

```
<div class = "container">
<div class = "row">
<div class = "col-*-*"></div>
<div class = "col-*-*"></div>
</div><div class = "row">...</div>
</div><div class = container">.....
```

18) What are the types of layouts we have in Bootstrap?

Two major types of layouts are there in Bootstrap-

- **Fixed Layout-** For a standard screen that is 940px, we use this layout.
- **Fluid Layout-** We use this layout to create an app that is 100% wider and covers the width of the screen.

Note- We can use both types of layouts to make the design responsive.

19) What is the process of creating an alert in Bootstrap?

Creating an alert in Bootstrap involves using predefined classes to style a notification box. These classes specify the alert's appearance, such as color and icon, effectively conveying messages.

20) How can you wrap content in Bootstrap?

You can use the command **".container"** to wrap and center the content. It provides fixed width responsive container.

- HTML

```
<div class="container">
  <h1>This is Ninja's page.</h1>
  <p>Coding Ninja is the best.</p>
</div>
```

21) How can we make an image round in Bootstrap 4?

".rounded" class is used to make an image rounded in Bootstrap.

For example -

- HTML

```

```

The image will look something like this -

22) What are responsive utility classes in Bootstrap?

For concealing or exhibiting the HTML elements, which are based on screen resolution and discerned by a media query, we use a set of classes called Responsive utility.

Example- "hidden-md-down"

23) What is the use of the carousel plugin in Bootstrap?

We use the carousels plugin in Bootstrap:

To make sliders on the website pages.

To display large contents within a small space by adding a slider.

Examples:

.carousel('options')

With optional options object, a carousel is initialized and starts cycling through items.

.carousel('next')

This type of carousel cycles to the next item. In other words, it is a slideshow for cycling through a series of information/content.

.carousel('cycle')

This type of carousel cycles through the items of carousel from left to right.

.carousel('pause')

This type of carousel stops the carousel from cycling through the carousel items.

.carousel('prev')

This type of carousel cycles to the previous item.

.carousel('number')

This cycles the carousel to a particular frame.

.carousel('dispose')

This one destroys a carousel of an element.

24) How can we draw a toolbar of buttons in Bootstrap?

Unique sets of button groups are combined by the class **.btn-toolbar** to create more complex components.

25) What is a button group in Bootstrap?

When you want to stack multiple buttons on a single line together, we use button groups. Also, when you want to place items in a group, like alignment buttons together.

We use **.btn-group** class for building a basic button group if we use the **.btn** type with the **.btn-group** class to wrap a series of buttons.

Bootstrap Interview Questions for Experienced

26) How can Bootstrap be used for making thumbnails?

You need to follow the below-given steps for creating a thumbnail using Bootstrap-

- With the class of **.thumbnail**, add **<a>**. the tag around an image.
- It will help to add four pixels of padding and a gray border.
- Now while hovering on to the image, an animated glow will outline the image.

27) How do you make images responsive?

A class called **.img-responsive** to an "a" tag to make it responsive for users. Other way is to add **.img-fluid** class to **** HTML tag to make image responsive. It applies **.max-width:100%**, **styles** and **height : auto**.

28) What are the types of lists that Bootstrap supports?

There are three types of lists supported by Bootstrap-

- **Ordered List**
Sequential order is followed by this type of list and is prefaced by numbers.
- **Unordered List**
This type of list doesn't have any particular order and is styled with bullets. If you don't want the bullets, then by using the class **.list-unstyled**, you can remove the styling.
- **Definition List**
<dt> and **<dd>** elements can be used to define each list item in this type of list. **<dt>** stands for definition term and **<dd>** is the definition of **<dt>** (definition term).

29) How can the Nav element be created in Bootstrap?

Bootstrap provides various options for styling navigation elements. Same markup and base class **.nav** are used by all of them. for creating a tabular navigation bar or tabs, you need to follow the below-given steps-

- You can start with an unordered list with a base class of **.nav**
- Now you can add **.nav-tabs** class
-

30) What is Bootstrap well?

It is a container that makes the content appear sunken. We can also use it to give an inset effect on our webpage. However, a developer can create and warp content in a well with the help of <div> tag in HTML and class .well. It makes the content looks recessed on your web page.

- HTML

```
<!DOCTYPE html>
<html lang="en">

<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
  <script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.6.0/jquery.min.js"></script>
  <script
src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></scri
pt>
</head>

<body>
  <div class="container">
    <h2>Coding Ninjas</h2>
    <div class="well">Ninja basic well</div>
  </div>
</body>

</html>
```

31) What do you mean by Bootstrap breadcrumb?

For displaying information for a site in the hierarchy, we use Bootstrap breadcrumbs. With a .breadcrumb class, it is an unordered list.

For example- in the case of blogs, breadcrumbs can be used to display categories, date of publishing, or tags. Here the separator is automatically added by CSS.

32) What are Glyphicons in Bootstrap? How do we use it?

In our web projects, we can use icon fonts called Glyphicons. Glyphicons halfings require licensing as they are not free. But they are free of cost for Bootstrap projects.

You must use the following code anywhere in your project to use the icons. You can leave space between the text and the icon for proper padding.

- HTML

```
<span class = "glyphicon glyphicon-man"></span>
```

33) What is 'normalize' in Bootstrap?

To ensure whether the content on your web page is consistent or not across different browsers, we use Normalize in Bootstrap. In other words, Bootstrap uses Normalize for establishing cross-browser consistency. It is a modern and HTML5-ready alternative to CSS resets. Better cross-browser consistency is provided by this small CSS file in the default styling of HTML elements.

34) What do you mean by contextual classes of the table in Bootstrap?

Class	Description
.warning	A warning is identified that may need attention.
.success	Positive or successful action is indicated here.
.danger	A potentially harmful or dangerous action is indicated here.
.active	To a particular row or cell, it applies the hover style/color.

35) What is the difference between Foundation and Bootstrap?

Bootstrap	Foundation
• An unlimited number of UI (User Interface) elements are offered/provided by Bootstrap. • Pixels are used by Bootstrap. • Here design is encouraged for both desktop and mobile. • LESS (Leaner Style Sheets) are supported by Bootstrap as it is a preprocessor.	• A minimal number of UI (User Interface) elements are provided here. • REMs (Risk Evaluation and Mitigation Strategy) are used by the foundation. • Mobile design is the priority of the foundation. • As the foundation is a preprocessor, it supports Sass (Syntactically Awesome Style Sheets) and Compass.

36) What do you mean by page header in Bootstrap?

A page header is used to add spacing around page headings, which can be created using the class .page-header.

37) Why do we use 'Jumbotron' in Bootstrap?

To highlight the web page's content, we use the jumbotron in Bootstrap. It provides a margin for the landing page's content and enlarges the heading. You must create a container div with the class .jumbotron to implement it.

38) What is the affix plugin in Bootstrap?

We are allowed to affix an <div> HTML tag to a particular location by affix plugin. We can even turn on and off the pinning by using this plugin, for example- social icons. It will start at one position, but when the web page reaches a specific point, the <div> tag will be frozen at a particular place and no longer scroll with the rest of the web page.

39) Discuss Bootstrap tables and various classes that can change the appearance of the table.

Bootstrap offers convenient tools for building and styling tables on web pages. It delivers different classes that permit you to modify the table's appearance. These classes can change borders, spacing, and color, making designing tables that match your website's style more comfortable.

40) What is the Bootstrap grid system? Give an example.

Bootstrap's grid system is a responsive, mobile-first system that can be scaled up to 12 columns per the increase in device size or viewport size. Predefined classes for easy layout options and powerful mix-ins for generating successful semantic layouts are featured in this system.

- HTML

```
<div class="container">
  <div class="row">
    <div class="col-sm">
      One of three columns
    </div>
    <div class="col-sm">
      One of three columns
    </div>
    <div class="col-sm">
      One of three columns
    </div>
  </div>
</div>
```

It will look something like this-

41) Explain the difference between Bootstrap's grid classes .container, .container-fluid, and .container-xl and when to use each one.

The **.container** class creates a fixed-width container with responsive padding on the sides, making it suitable for most layouts. On the other hand, **.container-fluid** creates a full-width container that spans the entire viewport width, making it useful for full-width designs or when you want content to extend to the edges. Additionally, the **.container-xl** class, introduced in Bootstrap 5, allows creating a container with an extended max-width for extra-large screens. It's useful for maintaining readability on larger displays without stretching content too wide.

42) What are Bootstrap utilities, and how do they differ from components? Provide examples of some commonly used Bootstrap utilities.

Bootstrap utilities are classes that provide quick and easy styling for common tasks like spacing, typography, and visibility. Unlike components, utilities are applied directly to HTML elements without the need for additional markup. For example, classes like **.m-1** for margin, **.text-center** for text alignment, **.d-none** for hiding elements, and **.bg-primary** for background color are commonly used utilities in Bootstrap.

43) Describe how you can customize Bootstrap components using Sass variables and mixins. Provide examples of how you might customize the default Bootstrap theme.

You can customize Bootstrap components using Sass variables and mixins. Sass variables allow you to override default values like colors, fonts, and spacing by simply reassigning them before importing Bootstrap's stylesheets. Mixins provide reusable CSS declarations that can be included in your custom stylesheets. For example, you can change the primary color by modifying the **\$primary** variable or create custom button styles using the **button-variant()** mixin.

44) Explain the concept of Bootstrap modals and how you can customize them to suit different use cases. Provide examples of how you might add custom animations or functionality to Bootstrap modals.

Bootstrap modals are dynamic dialog boxes that float above the rest of the content. You can customize modals by adding custom classes or modifying default styles. To add animations, you can use CSS transitions or JavaScript/jQuery to manipulate modal elements' visibility or properties. For example, you can use CSS animations to create fade-in/fade-out effects or jQuery to add custom functionality like closing the modal on clicking outside.

45) How do you handle browser compatibility issues when using Bootstrap? Discuss any specific considerations or techniques you might use to ensure consistent rendering across different browsers.

To handle browser compatibility issues when using Bootstrap, you should use Bootstrap's built-in responsive classes and grid system to create layouts that adapt to different screen sizes and devices. Test your website on multiple browsers and devices during development and use CSS prefixes or polyfills to address any compatibility issues. Additionally, consider using a CSS preprocessor like Autoprefixer to automatically add vendor prefixes to your CSS code and ensure consistent rendering across different browsers.

46) Discuss the advantages and disadvantages of using Bootstrap for building responsive websites compared to other CSS frameworks or writing custom CSS.

The advantages of using Bootstrap include faster development time, consistent styling across browsers, and built-in responsive features. However, disadvantages may include a larger file size, limitations in customization, and the risk of websites looking generic if not customized. While Bootstrap offers convenience and ease of use, some developers prefer writing custom CSS for more tailored and unique designs or using alternative CSS frameworks that better suit their project requirements.

47) Explain the Bootstrap grid system breakpoints and how you can use them to create responsive layouts for different screen sizes. Provide examples of how you might design a layout that adjusts seamlessly between desktop, tablet, and mobile devices.

Bootstrap grid system breakpoints include **sm**, **md**, **lg**, and **xl**, corresponding to small, medium, large, and extra-large screen sizes. You can use grid classes like **.col-md-6** to specify column widths at different breakpoints, ensuring your layout adapts to various screen sizes. For example, you can design a layout with multiple columns that stack vertically on smaller screens and horizontally on larger screens, ensuring seamless responsiveness across desktop, tablet, and mobile devices.

48) What are Bootstrap themes, and how do they work? Discuss the process of creating a custom Bootstrap theme and integrating it into your project.

Bootstrap themes are collections of stylesheets and assets that customize the appearance of Bootstrap components. To create a custom Bootstrap theme, you can override default Bootstrap variables or create custom CSS rules targeting specific components. Integrate the custom theme by including the theme's CSS file after the Bootstrap CSS file in your HTML. You can also use third-party Bootstrap theme generators or theme marketplaces to find and apply pre-made themes to your project.

49) Describe the process of creating a custom Bootstrap component from scratch. Discuss the steps involved, including HTML structure, CSS styling, and JavaScript functionality.

To create a custom Bootstrap component from scratch, start by defining the HTML structure for the component, ensuring it follows accessibility best practices. Then, style the component using CSS, applying Bootstrap classes or custom styles as needed. Finally, implement any necessary JavaScript functionality, such as event handling or dynamic content updates, to enhance the component's functionality and interactivity.

50) Discuss some advanced techniques for optimizing performance when using Bootstrap in large-scale projects. This may include strategies for minimizing CSS and JavaScript file sizes, lazy loading components, and optimizing images for faster page load times.

Advanced techniques for optimizing performance when using Bootstrap in large-scale projects include using tools like PurifyCSS or UnCSS to remove unused CSS classes from your stylesheets, minifying and concatenating CSS and JavaScript files to reduce file size and minimize HTTP requests, implementing lazy loading for components like images or JavaScript modules to defer loading until they're needed, and optimizing images by resizing, compressing, and using modern image formats like WebP to reduce file size without sacrificing quality. Additionally, consider using a content delivery network (CDN) to cache Bootstrap's assets and improve loading times for users.

Frequently Asked Questions

1. What is Bootstrap, and why is it used in web development?

Bootstrap is a popular open-source CSS framework used to build responsive and mobile-first websites quickly using pre-built components and utilities.

2. What are the main features of Bootstrap?

Key features include a responsive grid system, pre-styled components, JavaScript plugins, and strong browser compatibility.

3. What are the different types or versions of Bootstrap?

Bootstrap has evolved through versions like 3, 4, and 5. Each version introduces improvements in layout, components, and responsive design techniques.

4. What is the basic use of Bootstrap in front-end development?

Bootstrap simplifies the design process by offering ready-to-use HTML, CSS, and JavaScript components for creating responsive layouts.

5. What are the advantages of using Bootstrap in your projects?

It speeds up development, ensures responsiveness, supports consistent design, and integrates easily with other frameworks and tools.